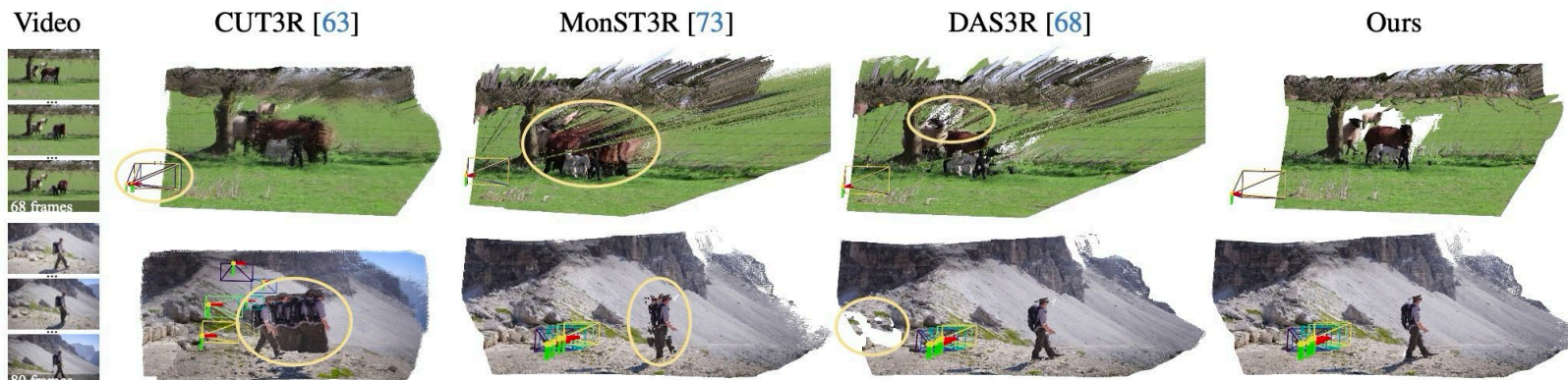
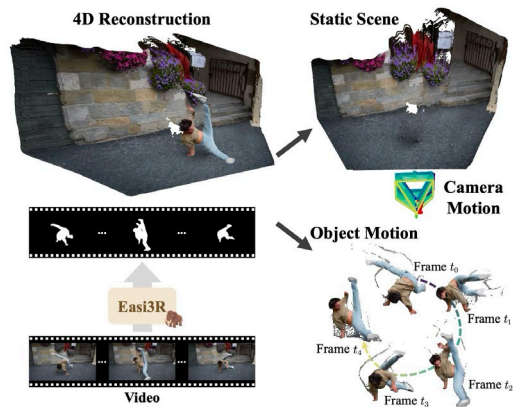




Easi3R: Estimating Disentangled Motion from DUS3R Without Training

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University of Tübingen, Tübingen AI Center⁴

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Presenter: 송우석

- **site:** <https://easi3r.github.io/>
- **github:** <https://github.com/Inception3D/Easi3R>
- **paper:** <https://arxiv.org/abs/2503.24391>

Index

- Problem & Motivation
- Method
- Experiments
- Conclusion

Recovery of geometry, motion

- foundation task of computer vision
 - downstream application: novel view synthesis, AR/VR, autonomous navigation, robotics etc
 - literature commonly identifies this problem as ‘SfM’
 - has been the core focus in 3D vision over decades
 - yielding **mature algorithms** that performs well under stationary conditions & wide baselines
- ⇒ **but these algorithms often fails on dynamic video input** (fail = degradation of accuracy & robustness)
- main reason: **object dynamics** (common component in real-world videos)
 - moving objects **violate fundamental assumptions of homography & epipolar consistency** in traditional SfM methods
 - dynamic video
 - entangled camera & object motion → **hard to disentangle these two motions**
 - motion with rich texture → **degrades camera pose estimate performance**

Related works

• SfM & SLAM

- long foundation for **3D structure** and **camera pose estimation**
- refine structure and motion estimates by
 - associating 2D correspondences or minimizing photometric errors
 - bundle adjustment (BA)

⇒ effective with dense input but struggle with limited camera parallax, ill-posed conditions

- DUST3R(+ follow-up methods) overcame above problems, but limited on static scenes

• Pose-free Dynamic Scene Reconstruction

- SLAM + dynamic scenes
 - semantic segmentation, optical flows to enhance SLAM's resilience in dynamic scenes

• Another line of work

- stable video depth estimation (+ geometric constraints, generative priors)
- optimization-based methods

- **CasualSAM**: fine-tune depth network at test time using pre-computed optical flow
- **Robust-CVD**: refines pre-computed depth, camera pose by leveraging masked optical flow
- **MegaSaM**: incorporate DROID-SLAM, optical flow, depth initializations

- point-map-based methods

- **MonST3R**: fine-tuned DUST3R with dynamic datasets and incorporate optical flow
- **DAS3R**: trains DPT on top of MonST3R and performs feedforward segmentation estimation
- **CUT3R**: encode scene into persistent state and performs feedforward 3D reconstruction

Easi3R ✓

- training-free, plug-in-play adaption
- no fine-tuning, almost no additional cost
- scalable, efficient alternative for handling real-world dynamic videos

?

mation. CUT3R [63] fine-tunes MAST3R [24] on both static and dynamic datasets, achieving feedforward reconstruction.

⇒ but these methods needs costly training on diverse motion patterns to generalize well

Related works

- Motion Segmentation

- Classical approaches

- rely on optical flow estimation, point tracking
 - only trained in 2D datasets

⇒ struggle with occlusions, disentangling object, camera motion

- robustness enhanced approaches

- **RoMo**: incorporate epipolar geometry, accurate calibration with COLMAP

⇒ focus primarily on removing dynamic objects and reconstructing static scene elements only

- **SAM2**: disambiguate object, camera motion

- complete 4D reconstruction

- **MonST3R**: fine-tuned DUStr with dynamic datasets and incorporate optical flow

- **DAS3R**: trains DPT on top of MonST3R and performs feedforward segmentation estimation

Key Insight

⇒ dynamic segmentation can be extracted from pre-trained 3D reconstruction models like DUStr

Easi3R ✓

- a simple, yet robust strategy to isolate this information from the attention layers
 - no need for optical flow or pre-training on segmentation datasets

Easi3R

- Goal

input

- $\{I^t\}_{t=1}^T$: video sequence

output

- M^t : object motion
- P^t, K^t : camera movement
- + X^t : point map

- Process

1. how the [DUST3R](#) model handles videos
2. mechanisms of [attention aggregation](#) in spatial and temporal dimensions
3. how aggregated cross-attention maps can be leveraged to decompose dynamic object segmentation ([re-weighting](#))

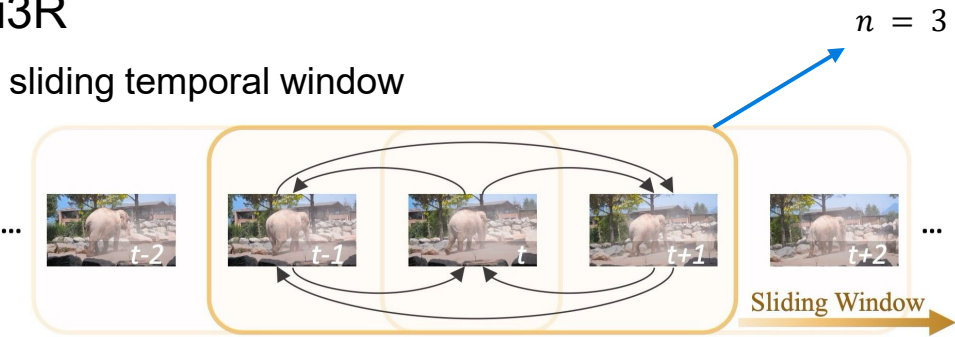
• DUST3R

$$X^{a \rightarrow a}, X^{b \rightarrow a} = \text{DUST3R}(I^a, I^b) \quad (1)$$

- $X^{b \rightarrow a}$: point map of input I^b predicted in the view a coordinate space
- In multi-views, DUST3R globally aligns the pairwise predictions into a joint coordinate space using a connectivity graph
 ⇒ computational redundancy problem (view connectivity is know for video sequence) (= waste)

• Easi3R

- sliding temporal window



- pair set

$$\varepsilon^t = \{(a, b) | a, b \in [t - \frac{n-1}{2}, \dots, t + \frac{n-1}{2}], a \neq b\}$$

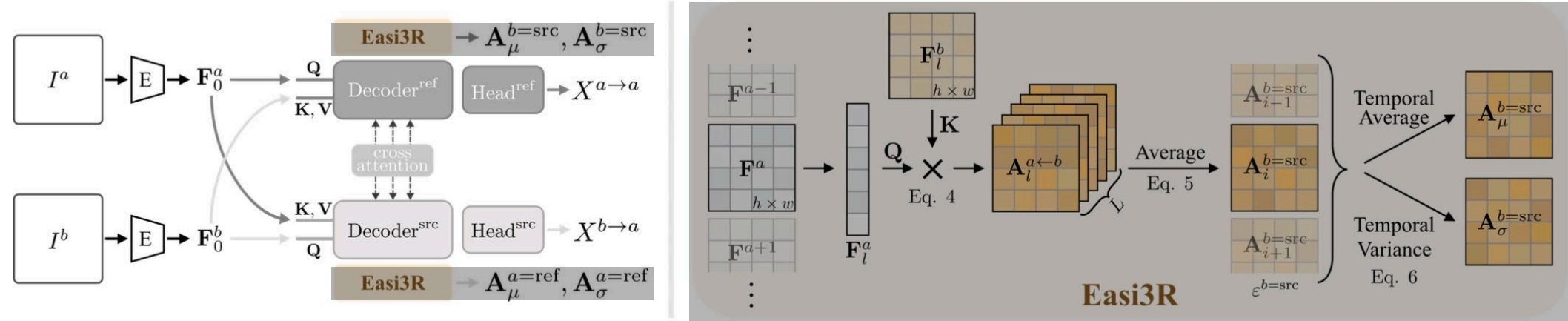
$$\mathcal{X}^* = \arg \min_{\mathcal{X}, \mathbf{P}, \mathbf{s}} \sum_{t \in T} \sum_{i \in \varepsilon^t} \|\mathcal{X}^a - \mathbf{s}_i^t \mathbf{P}_i^t X^{a \rightarrow a}\|_1 + \|\mathcal{X}^b - \mathbf{s}_i^t \mathbf{P}_i^t X^{b \rightarrow a}\|_1 \quad (2)$$

global alignment of pair-wise prediction



DUST3R can't handle frame with dynamic object

• Secret behind DUST3R



• 2 branches

- I^a : reference image
- I^b : source image

• process

1. encode image into token representation F with weight-sharing ViT encoder
2. exchange information both within and between views (self-attention + [cross-attention](#))
 - use previous block ($l - 1$)
3. point map prediction with 2 heads using feature tokens
4. iterate minimizing Euclidean distance between predicted point map and GT point map

$l = 1, \dots, L$
block index

$$\begin{aligned} F_0^a &= \text{Encoder}(I^a) \\ F_0^b &= \text{Encoder}(I^b) \end{aligned}$$

self-attention (within views) + [cross-attention](#) (between views)

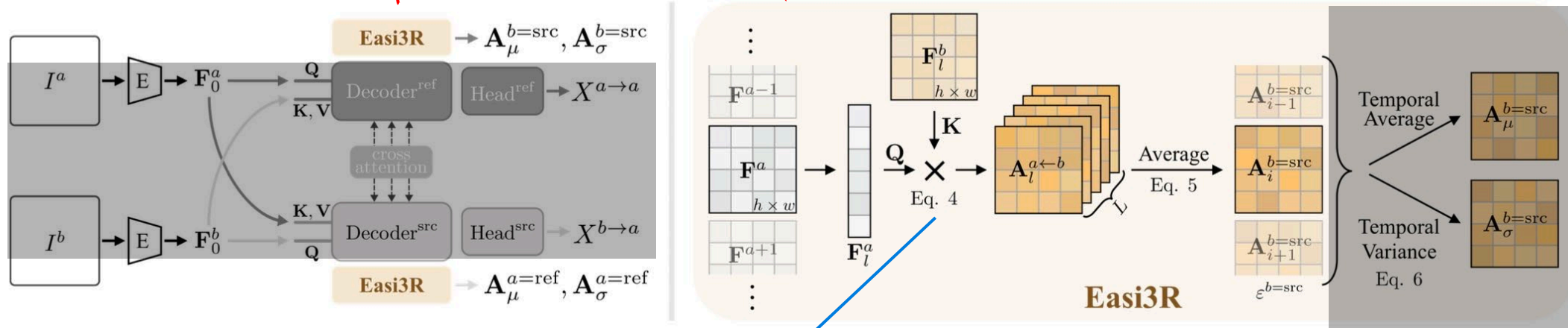
$$\begin{aligned} F_l^a &= \text{DecoderBlock}_l^{\text{ref}}(F_{l-1}^a, F_{l-1}^b) \\ F_l^b &= \text{DecoderBlock}_l^{\text{src}}(F_{l-1}^b, F_{l-1}^a) \end{aligned} \tag{3}$$
$$\begin{aligned} X^{a \rightarrow a} &= \text{Head}^{\text{ref}}(F_0^a, \dots, F_L^a) \\ X^{b \rightarrow a} &= \text{Head}^{\text{src}}(F_0^b, \dots, F_L^b) \end{aligned}$$

point map prediction

Easi3R ✓

- DUST3R [implicitly learns rigid view transformations](#) with [cross-attention](#) layer, [assigning low attention values to tokens that violate epipolar geometry constraints](#) (such as texture-less, under-observed, and dynamic regions)
- aggregating cross-attention outputs across spatial & temporal dimensions → [enabling motion extraction](#)

- Easi3R



- trainable linear function (projection function of Q, K)

$$\ell_{\mathbf{Q}}(\cdot), \ell_{\mathbf{K}}(\cdot) \quad \mathbf{Q} = \ell_{\mathbf{Q}}(\mathbf{F}) \in \mathbb{R}^{(h \times w) \times c}$$

- attention map

$$\mathbf{A}_l^{a \leftarrow b} = \mathbf{Q}_l^a \mathbf{K}_l^{bT} / \sqrt{c}, \quad \mathbf{A}_l^{b \leftarrow a} = \mathbf{Q}_l^b \mathbf{K}_l^{aT} / \sqrt{c} \quad (4)$$

$$\mathbf{A}_l^{a \leftarrow b}, \mathbf{A}_l^{b \leftarrow a} \in \mathbb{R}^{(h \times w) \times h \times w}$$

$\mathbf{A}_l^{a \leftarrow b}$ determines how the information is aggregated from the view b into the view a in the l th decoder block

$$\mathbf{V} = \ell_{\mathbf{V}}(\mathbf{F}) \in \mathbb{R}^{(h \times w) \times c}$$

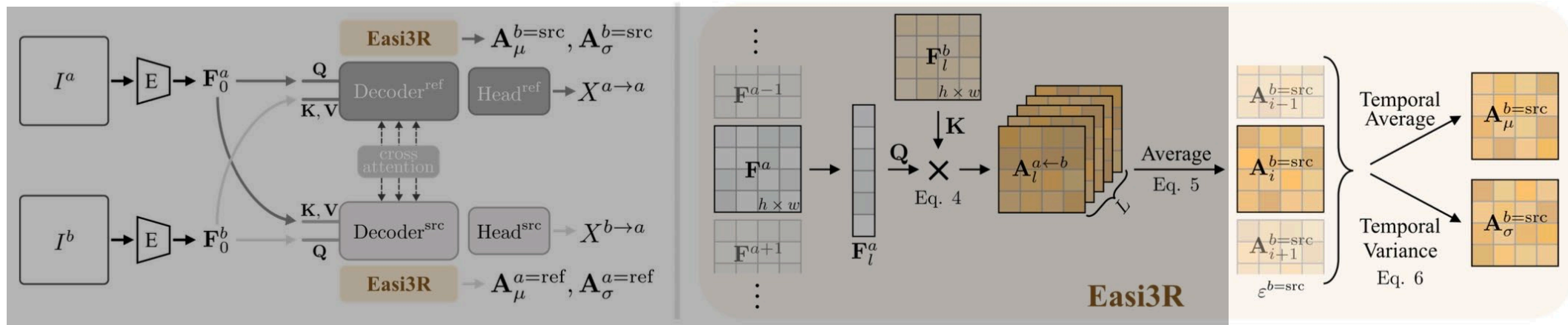
$$\text{softmax}(\mathbf{A}_l^{a \leftarrow b}) \mathbf{V}^b$$

- average attention map

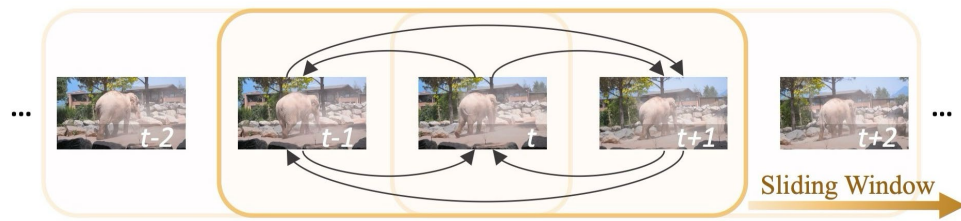
$$\begin{aligned} \mathbf{A}^{b=src} &= \sum_l \sum_x \mathbf{A}_l^{a \leftarrow b}(x, y, z) / (L \times h \times w) \\ \mathbf{A}^{a=ref} &= \sum_l \sum_x \mathbf{A}_l^{b \leftarrow a}(x, y, z) / (L \times h \times w) \end{aligned} \quad (5)$$

average attention map captures the overall influence of tokens from one view to another across all decoder layers

• Easi3R



• Temporal attention maps



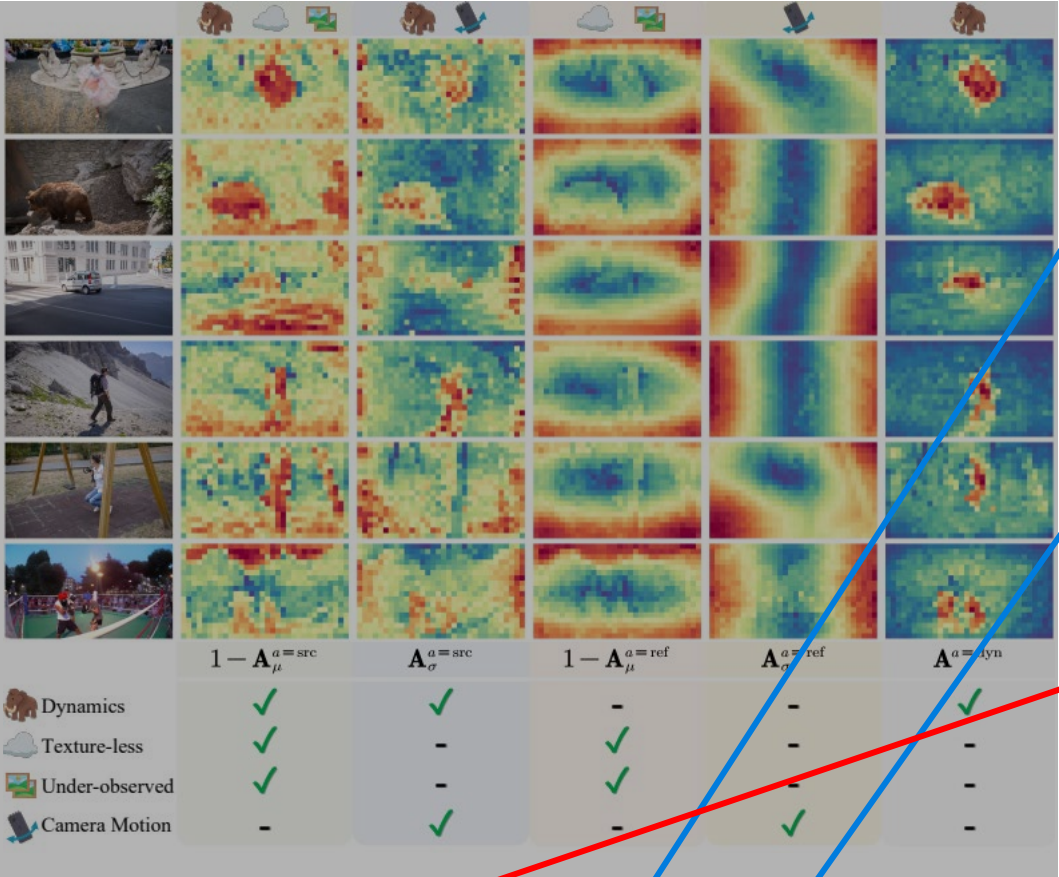
- $2(n-1)$ attention maps per frame

$$\mathbf{A}_{\mu}^{b=\text{src}} = \text{Mean}(\mathbf{A}_i^{b=\text{src}}), \mathbf{A}_{\sigma}^{b=\text{src}} = \text{Std}(\mathbf{A}_i^{b=\text{src}})$$

$$(6) \quad \begin{aligned} & i \in \varepsilon^{b=\text{src}} \\ & \varepsilon^{b=\text{src}} = \{(a, b) | \text{src} = b, a \in [t-n, \dots, t+n], a \neq b\} \end{aligned} \quad (7)$$

$$(6) \quad \begin{aligned} & i \in \varepsilon^{b=\text{ref}} \\ & \varepsilon^{b=\text{ref}} = \{(a, b) | \text{ref} = b, a \in [t-n, \dots, t+n], a \neq b\} \end{aligned} \quad (8)$$

Easi3R



Temporal attention maps

$1 - \mathbf{A}_\mu^{a=\text{ref}}$

$\mathbf{A}_\mu^{a=\text{ref}}$

- smooth attention values
- low attention values with texture-less regions, under-observed areas
 $\Rightarrow$ DUS3R believes that they are less useful for registration
 $\Rightarrow$ can be extracted using $(1 - \mathbf{A}_\mu^{a=\text{ref}})$

$\mathbf{A}_\sigma^{a=\text{ref}}$

- changes of token contribution in image coordinate space
- pixels perpendicular to the direction of motion generally share similar pixel flow speeds
 $\Rightarrow$ consistent deviation that allow to infer camera motion

$1 - \mathbf{A}_\mu^{a=\text{src}}$

$\mathbf{A}_\mu^{a=\text{src}}$

- indicates low texture & under-observed area + dynamic object
 $\Rightarrow$ they violate the rigid body transformation prior that DUS3R has learned from the 3D dataset
 $\Rightarrow$ can be extracted using $(1 - \mathbf{A}_\mu^{a=\text{src}})$

$\mathbf{A}_\sigma^{a=\text{src}}$

- highlights camera & object motion
 $\Rightarrow$ these areas continuously changes over time, leading high deviation

$$\mathbf{A}^{a=\text{dyn}} = (1 - \mathbf{A}_\mu^{a=\text{src}}) \cdot \mathbf{A}_\sigma^{a=\text{src}} \cdot \mathbf{A}_\mu^{a=\text{ref}} \cdot (1 - \mathbf{A}_\sigma^{a=\text{ref}}) \quad (9)$$

$$\mathbf{M}^t = [\mathbf{A}^{t=\text{dyn}} > \alpha] \quad (\mathbf{M}^t \in \mathbb{R}^{h \times w}) \quad \rightarrow : \text{inverted}$$

$\Rightarrow$ no worries for o.o.d input (e.g. black pixels) resulting performance degradation

(supplementary material)

• Clustering Fusion

$$\bar{\mathbf{F}} = [\mathbf{F}_0^1; \mathbf{F}_0^2; \dots; \mathbf{F}_0^T] \in \mathbb{R}^{(T \times h \times w) \times c} \quad (12)$$

number of cluster ($k = 64$)

$$C = \text{KMeans}(\bar{\mathbf{F}}, k), \quad C^t(x, y) \in \{1, \dots, k\}, \quad \forall t, x, y \quad (13)$$

dynamic score

$$s_c = \frac{\sum_t \sum_{i,j} \mathbb{1}[C^t(x, y) = c] \cdot \mathbf{A}^{t=\text{dyn}}(x, y)}{\sum_t \sum_{x,y} \mathbb{1}[C^t(x, y) = c]} \quad (14)$$

indicator function

cluster fused dynamic attention map

$$\mathbf{A}_{\text{fuse}}^{t=\text{dyn}}(x, y) = s_{C^t(x,y)} \quad (15)$$

$$\mathbf{M}^t(x, y) = \mathbb{1}[\mathbf{A}_{\text{fuse}}^{t=\text{dyn}}(x, y) > \alpha] \quad (16)$$

- α is automatic image thresholding using Otsu's method

Backbone	Ablation	DAVIS-16		DAVIS-17		DAVIS-all	
		JM↑	JR↑	JM↑	JR↑	JM↑	JR↑
DUS3R	w/o $\mathbf{A}_{\mu}^{a=\text{src}}$	45.1	45.2	42.8	39.9	42.2	38.5
	w/o $\mathbf{A}_{\sigma}^{a=\text{src}}$	42.3	50.0	35.0	37.0	30.9	28.3
	w/o $\mathbf{A}_{\mu}^{a=\text{ref}}$	33.3	28.4	31.5	27.9	32.5	29.7
	w/o $\mathbf{A}_{\sigma}^{a=\text{ref}}$	47.7	54.1	46.2	54.3	43.7	48.6
	w/o Clustering	40.0	38.5	38.3	38.3	34.3	30.5
	Full	53.1	60.4	49.0	56.4	44.5	49.6
MonST3R	w/o $\mathbf{A}_{\mu}^{a=\text{src}}$	47.2	51.5	44.4	46.7	40.9	41.5
	w/o $\mathbf{A}_{\sigma}^{a=\text{src}}$	49.7	60.1	48.7	57.8	44.9	49.6
	w/o $\mathbf{A}_{\mu}^{a=\text{ref}}$	46.4	54.0	47.4	55.9	45.3	50.7
	w/o $\mathbf{A}_{\sigma}^{a=\text{ref}}$	50.7	62.6	51.0	60.2	50.3	56.8
	w/o Clustering	45.5	46.7	45.3	48.1	42.1	43.5
	Full	57.7	71.6	56.5	68.6	53.0	63.4

• Attention re-weighting

$$\mathbf{M}^{a \leftarrow b} = (1 - \mathbf{M}^a) \otimes \mathbf{M}^{bT}$$

static attention mask of view a

dynamic attention mask of view b

$$\text{softmax}(\tilde{\mathbf{A}}_l^{a \leftarrow b}) = \begin{cases} 0 & \text{if } \mathbf{M}^{a \leftarrow b} \\ \text{softmax}(\mathbf{A}_l^{a \leftarrow b}) & \text{otherwise} \end{cases} \quad (10)$$

- resulting tokens from [dynamic regions in view \$b\$](#) that **do not contribute** to [static regions in view \$a\$](#)
- re-weighting only applied on [reference view decoder](#) (source view requires a static reference)

Backbone	Re-weighting	Flow-GA	Pose Estimation			Reconstruction					
			ATE↓	RTE↓	RRE↓	Accuracy↓		Completeness↓		Distance↓	
						Mean	Median	Mean	Median	Mean	Median
DUS3R	Ref + Src	✗	0.030	0.026	1.777	0.775	0.596	1.848	0.778	0.342	0.224
	Ref	✗	0.029	0.025	1.774	0.772	0.596	1.813	0.757	0.336	0.219
	Ref	w/o Mask	0.026	0.017	1.472	0.940	0.831	1.654	0.685	0.336	0.220
	Ref	w/ Mask	0.021	0.014	1.092	0.703	0.589	1.474	0.586	0.301	0.186
MonST3R	Ref + Src	✗	0.040	0.032	1.751	0.848	0.744	1.850	1.003	0.398	0.292
	Ref	✗	0.038	0.032	1.736	0.846	0.660	1.840	0.983	0.390	0.290
	Ref	w/o Mask	0.033	0.023	1.495	0.969	0.796	1.752	0.998	0.368	0.273
	Ref	w/ Mask	0.030	0.021	1.390	0.834	0.643	1.661	0.916	0.350	0.255

ablation study of Camera Pose Estimation and Point Cloud Reconstruction

Global Alignment

$$\mathcal{X}^* = \arg \min_{\mathcal{X}, \mathbf{P}, \mathbf{s}} \sum_{t \in T} \sum_{i \in \mathcal{E}^t} \|\mathcal{X}^a - \mathbf{s}_i^t \mathbf{P}_i^t X^{a \rightarrow a}\|_1 + \|\mathcal{X}^b - \mathbf{s}_i^t \mathbf{P}_i^t X^{b \rightarrow a}\|_1 \quad (2)$$

global alignment of pair-wise prediction

$$\mathcal{X}^b : (\mathbf{P}^a, \mathbf{K}^a) \rightarrow (\mathbf{P}^b, \mathbf{K}^b)$$

computed optical flow

estimated optical flow

$$\mathcal{L}_{\text{flow}} = \sum_{t \in T} \sum_{i \in \mathcal{E}^t} (1 - \mathbf{M}^a) \cdot \|\hat{\mathcal{F}}_i^{a \rightarrow b} - \mathcal{F}_i^{a \rightarrow b}\|_1 + (1 - \mathbf{M}^b) \cdot \|\hat{\mathcal{F}}_i^{b \rightarrow a} - \mathcal{F}_i^{b \rightarrow a}\|_1 \quad (11)$$

flow loss of static area

incorporation can achieve more robust outputs in terms of global pointmaps and pose sequence

but, optionally used (for fair comparison)

(segmentation)

Dynamic Object Motion

• dataset (video object segmentation benchmark)

- DAVIS-16
- DAVIS-17
- DAVIS-all

• metric

- JM (IoU mean)
- JR (IoU recall)

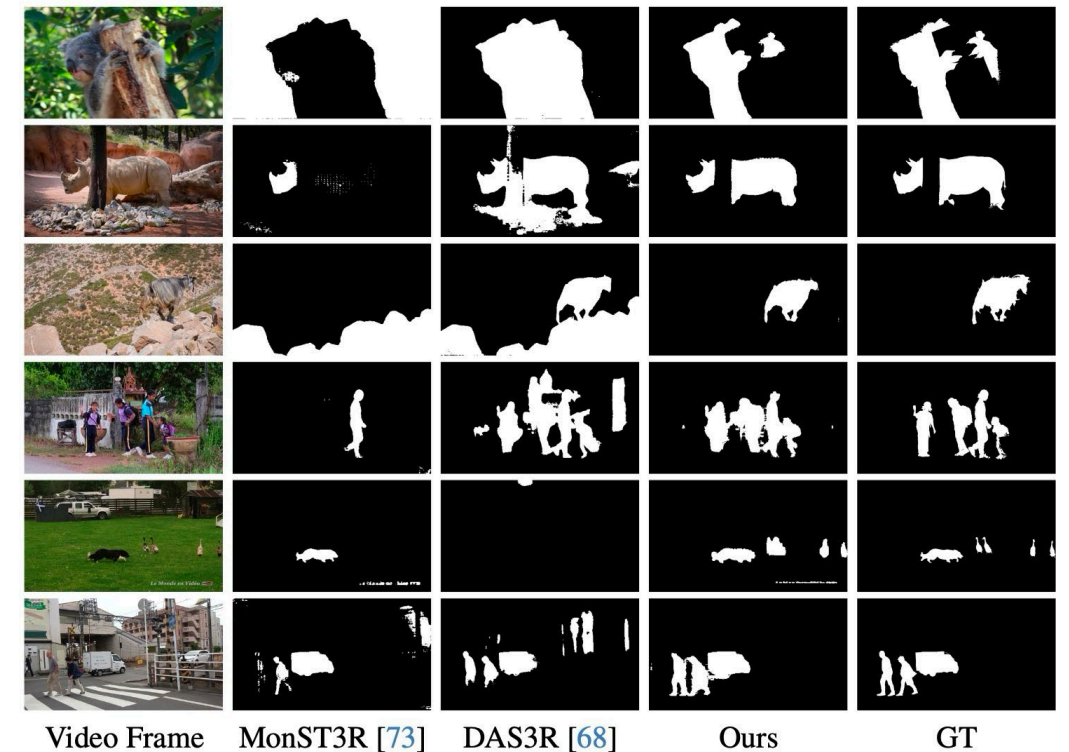
• result

- improved segmentation without need for flow
- even surpassed DAS3R (trained on dynamic mask labels)

+ flow-guided segmentation

outputs serve as prompt for SAM2

Method	Flow	DAVIS-16				DAVIS-17				DAVIS-all			
		w/o SAM2		w/ SAM2		w/o SAM2		w/ SAM2		w/o SAM2		w/ SAM2	
		JM↑	JR↑	JM↑	JR↑	JM↑	JR↑	JM↑	JR↑	JM↑	JR↑	JM↑	JR↑
DUST3R [64]	✓	42.1	45.7	58.5	63.4	35.2	35.3	48.7	50.2	35.9	34.0	47.6	48.7
MonST3R [73]	✓	40.9	42.2	64.3	<u>73.3</u>	38.6	38.2	56.4	59.6	36.7	34.3	51.9	54.1
DAS3R [68]	✗	41.6	39.0	54.2	55.8	43.5	42.1	57.4	61.3	43.4	38.7	53.9	54.8
Easi3R _{dust3r}	✗	53.1	60.4	67.9	71.4	49.0	<u>56.4</u>	<u>60.1</u>	<u>65.3</u>	<u>44.5</u>	<u>49.6</u>	<u>54.7</u>	<u>60.6</u>
Easi3R _{monst3r}	✗	57.7	71.6	70.7	79.9	56.5	68.6	67.9	76.1	53.0	63.4	63.1	72.6



(pose estimation)

Camera Motion

• dataset (dynamic benchmark dataset)

- DyCheck
 - diverse, in-the-wild dynamic videos captured from handheld cameras
- TUM-dynamics
 - major dynamic objects in relatively simple indoor scenarios
- ADT
 - egocentric videos, which are out-of-distribution for DUST3R's training set

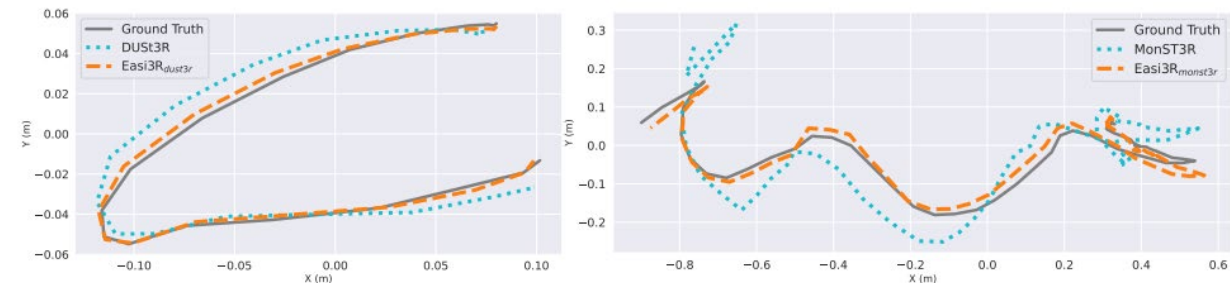
• metric

- ATE (Absolute Trajectory Error)
 - RTE (Relative Translation Error)
 - RRE (Relative Rotation Error)
- ⇒ after applying the Sim(3) alignment on the estimated camera trajectory to GT

• result

- best overall performance among all methods

Method	Flow	DyCheck			ADT			TUM-dynamics		
		ATE ↓	RTE ↓	RRE ↓	ATE ↓	RTE ↓	RRE ↓	ATE ↓	RTE ↓	RRE ↓
DUST3R [64]	✗	0.035	0.030	2.323	0.042	0.025	1.212	0.100	0.087	2.692
Easi3R_{dust3r}	✗	<u>0.029</u>	<u>0.025</u>	<u>1.774</u>	0.040	<u>0.021</u>	<u>0.880</u>	0.093	0.076	<u>2.366</u>
DUST3R [64]	✓	<u>0.029</u>	<u>0.021</u>	1.875	0.076	0.030	0.974	<u>0.071</u>	<u>0.067</u>	3.711
Easi3R_{dust3r}	✓	0.021	0.014	1.092	<u>0.042</u>	0.015	0.655	0.070	0.061	2.361
MonST3R [73]	✗	0.040	0.034	1.820	<u>0.045</u>	<u>0.024</u>	0.759	0.183	0.148	6.985
Easi3R_{monst3r}	✗	0.038	0.032	1.736	<u>0.045</u>	<u>0.024</u>	<u>0.715</u>	0.184	<u>0.149</u>	<u>6.311</u>
MonST3R [73]	✓	<u>0.033</u>	<u>0.024</u>	<u>1.501</u>	0.055	0.025	0.776	<u>0.170</u>	0.155	6.455
Easi3R_{monst3r}	✓	0.030	0.021	1.390	0.039	0.016	0.640	0.168	0.150	5.925



4D Reconstruction

dataset

- DyCheck

metric

- accuracy
 - nearest Euclidean distance from a reconstructed point to GT
- completeness
 - reverse of accuracy
- distance
 - Euclidean distance from a reconstructed point to GT

result

- Quantitative – more accurate reconstruction than other baselines, even comparable to CUT3R (concurrent & trained with extensive datasets)
- Qualitative – other baselines struggle with misalignment and entablement of dynamic & static reconstructions
⇒ resulting broken geometry, distortions, ghosting artifacts

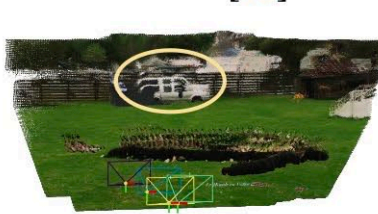
Method	Flow	Accuracy↓		Completeness↓		Distance↓	
		Mean	Median	Mean	Median	Mean	Median
DUST3R [64]	✗	0.802	<u>0.595</u>	1.950	0.815	0.353	0.233
Easi3R _{dust3r}	✗	0.772	0.596	1.813	0.757	0.336	0.219
DUST3R [64]	✓	0.738	0.599	<u>1.669</u>	<u>0.678</u>	0.313	0.196
Easi3R _{dust3r}	✓	0.703	0.589	1.474	0.586	0.301	0.186
MonST3R [73]	✗	0.855	0.693	1.916	1.035	0.398	0.295
Easi3R _{monst3r}	✗	0.846	0.660	1.840	0.983	0.390	0.290
MonST3R [73]	✓	0.851	0.689	<u>1.734</u>	<u>0.958</u>	<u>0.353</u>	0.254
Easi3R _{monst3r}	✓	0.834	0.643	1.661	0.916	0.350	<u>0.255</u>

Method	Flow	Accuracy↓		Completeness↓		Distance↓	
		Mean	Median	Mean	Median	Mean	Median
DUST3R [64]	✗	0.802	0.595	1.950	0.815	0.353	0.233
CUT3R [63]	✗	0.458	0.342	<u>1.633</u>	<u>0.792</u>	<u>0.326</u>	<u>0.229</u>
MonST3R [73]	✓	0.851	0.689	1.734	0.958	0.353	0.254
DAS3R [68]	✓	1.772	1.438	2.503	1.548	0.475	0.352
Easi3R _{monst3r}	✓	0.834	0.643	1.661	0.916	0.350	0.255
Easi3R _{dust3r}	✓	<u>0.703</u>	<u>0.589</u>	1.474	0.586	0.301	0.186

Video



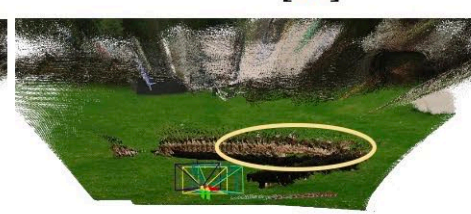
CUT3R [63]



MonST3R [73]



DAS3R [68]



Ours



Ablation study

Backbone	Re-weighting	Flow-GA	Pose Estimation			Reconstruction					
			ATE↓	RTE↓	RRE↓	Accuracy↓		Completeness↓		Distance↓	
						Mean	Median	Mean	Median	Mean	Median
DUS3R	Ref + Src	✗	0.030	0.026	1.777	0.775	0.596	1.848	0.778	0.342	0.224
	Ref	✗	0.029	0.025	1.774	0.772	0.596	1.813	0.757	0.336	0.219
	Ref	w/o Mask	0.026	0.017	1.472	0.940	0.831	1.654	0.685	0.336	0.220
	Ref	w/ Mask	0.021	0.014	1.092	0.703	0.589	1.474	0.586	0.301	0.186
MonST3R	Ref + Src	✗	0.040	0.032	1.751	0.848	0.744	1.850	1.003	0.398	0.292
	Ref	✗	0.038	0.032	1.736	0.846	0.660	1.840	0.983	0.390	0.290
	Ref	w/o Mask	0.033	0.023	1.495	0.969	0.796	1.752	0.998	0.368	0.273
	Ref	w/ Mask	0.030	0.021	1.390	0.834	0.643	1.661	0.916	0.350	0.255

Ablation	Variants	DAVIS-16		DAVIS-17		DAVIS-all	
		JM↑	JR↑	JM↑	JR↑	JM↑	JR↑
Window Size	3	76.0	89.2	70.8	82.4	65.7	76.2
	5*	70.7	79.9	67.9	76.1	63.1	72.6
	7	66.9	76.9	63.9	73.3	61.0	68.8
Number of Clusters	16	67.4	79.1	64.6	73.7	60.7	65.2
	32	71.6	83.9	68.0	78.3	65.1	75.2
	64*	70.7	79.9	67.9	76.1	63.1	72.6
	128	69.9	79.9	66.3	76.2	62.9	73.1
Thresholding Values	0.5	61.6	64.5	61.0	65.2	61.6	67.8
	0.7	70.2	85.0	62.8	71.8	58.1	67.0
	Otsu's method*	70.7	79.9	67.9	76.1	63.1	72.6

Backbone	Ablation	DAVIS-16		DAVIS-17		DAVIS-all	
		JM↑	JR↑	JM↑	JR↑	JM↑	JR↑
DUS3R	w/o $A_{\mu}^{a=src}$	45.1	45.2	42.8	39.9	42.2	38.5
	w/o $A_{\sigma}^{a=src}$	42.3	50.0	35.0	37.0	30.9	28.3
	w/o $A_{\mu}^{a=ref}$	33.3	28.4	31.5	27.9	32.5	29.7
	w/o $A_{\sigma}^{a=ref}$	47.7	54.1	46.2	54.3	43.7	48.6
	w/o Clustering	40.0	38.5	38.3	38.3	34.3	30.5
	Full	53.1	60.4	49.0	56.4	44.5	49.6
MonST3R	w/o $A_{\mu}^{a=src}$	47.2	51.5	44.4	46.7	40.9	41.5
	w/o $A_{\sigma}^{a=src}$	49.7	60.1	48.7	57.8	44.9	49.6
	w/o $A_{\mu}^{a=ref}$	46.4	54.0	47.4	55.9	45.3	50.7
	w/o $A_{\sigma}^{a=ref}$	50.7	62.6	51.0	60.2	50.3	56.8
	w/o Clustering	45.5	46.7	45.3	48.1	42.1	43.5
	Full	57.7	71.6	56.5	68.6	53.0	63.4

Conclusion

- **Easi3R**

- spatial & temporal attention mechanism behind DUS3R
- second inference pass (attention re-weighting)
- achieve training-free, robust 4D reconstruction
- outperforms SoTA methods in most cases

- Limitation

- fail when DUS3R/MonST3R backbones produce inaccurate depth predictions